

Let $P(x)$ be a good polynomial. Since P is monic of degree 7 with integer coefficients the leading coefficient is 1 so $|1| = 1$ is a root of P . Thus $P(1) = 0$ and the sum of all coefficients of P is zero. Let the coefficients of P be $a_7, a_6, \dots, a_0$ where $a_7 = 1$. Each $|a_i|$ is an odd positive integer strictly less than 2025^2 and each $|a_i|$ divides the constant term a_0 by the Rational Root Theorem. Let $D = |a_0|$. All $|a_i|$ divide D and D is the maximum of the $|a_i|$.

Claim 1: $D = 1$. Suppose $D > 1$. Let p be the smallest prime divisor of D . Since p divides D and no prime smaller than p divides any $|a_i|$ there exists a coefficient a_j with $|a_j| = p$. Evaluating $P(p) = 0$ yields:

$$p^7 + a_6 p^6 + \dots + a_0 = 0.$$

To cancel the leading term p^7 the coefficient a_6 must satisfy $a_6 = -p$ making the first two terms vanish: $p^7 - p \cdot p^6 = 0$. This forces $a_6 = -p$. Continuing recursively all coefficients from a_6 to a_1 must be $\pm p$. The constant term a_0 must then be $\pm p$ but $|a_0| = D$ so $D = p$. The sum of coefficients is:

$$1 + (-p) + \sum_{k=0}^5 (\pm p) = 0.$$

Let m be the net count of $+p$ terms among the six coefficients a_5 to a_0 . Then:

$$1 - p + p \cdot m = 0 \implies p(m - 1) = -1.$$

This equation has no integer solutions since $p \geq 3$ and the left side is a multiple of p while the right side is -1 . Contradiction. Thus $D \neq p$ and the assumption $D > 1$ is false. Hence $D = 1$.

Claim 2: All coefficients satisfy $|a_i| = 1$. Since $D = 1$ each $|a_i|$ divides 1 so $|a_i| = 1$. Thus every coefficient is ± 1 .

Claim 3: The sum of coefficients equals zero. The sum $1 + a_6 + \dots + a_0 = 0$ implies:

$$a_6 + \dots + a_0 = -1.$$

There are 7 coefficients a_6 to a_0 each ± 1 . Let k be the number of $+1$ s among them. Then:

$$k - (7 - k) = -1 \implies 2k - 7 = -1 \implies k = 3.$$

Thus exactly three of these coefficients are $+1$ and four are -1 .

Claim 4: The number of such polynomials is $\binom{7}{3}$. The coefficients a_6 to a_0 can be chosen by selecting 3 positions out of 7 to assign $+1$ with the rest -1 . The number of ways is:

$$\binom{7}{3} = \frac{7!}{3!4!} = 35.$$

Conclusion: All good polynomials must satisfy the conditions with $D = 1$ leading to exactly 35 distinct polynomials. No other polynomials exist as shown by the contradictions for $D > 1$. Therefore the number of good polynomials is 35.